

What's New In The Latest MICROCONTROLLERS



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Targeting the simple application tier, microcontroller (MCU) vendors continue to drive their products to break new limits of cost, current drain and size, while adding features to ease development, hoping to entice embedded system developers. What do the latest MCUs offer this year? Let us take a look.

IoT-targeted MCUs

"New or established, every semiconductor vendor needs a new paradigm to solve four critical issues in the Internet of Things (IoT), namely, design time, differentiation, time to revenue and design cost," explains Alexandru Voica, senior technology specialist, Imagination Technologies, in an interview with EFY. He adds that, in the IoT market, consumers have high expectations; devices need to be affordable, power-efficient and must work out-of-the-box.

IoT-targeted MCUs look mostly at network connectivity in the form of wired (Ethernet) or wireless (Wi-Fi, Bluetooth or others), lower power and a wide range of sensors like temperature, motion or humidity.

T. Anand, managing director, Kneutron, says, "I would bet mostly on addition of communication standards like Wi-Fi and Bluetooth Low Energy (BLE). An upcoming chip from Espressif Systems has both Wi-Fi and BLE put together in

one system on chip (SoC), ESP32, which could be another game-changer."

Ingenic M200 is another good example of an innovative SoC built from the ground up for wearable devices. It implements a power-saving hardware architecture where a high-performance microprocessor without interlocked pipeline stages, having a central processing unit (CPU) clocked at 1.2GHz, tackles most of the heavy-lifting, while less-demanding tasks are handled by a secondary low-power 300MHz MIPS CPU.

Besides the usual migration to higher-density technology node (90nm and beyond) to deliver more features, last year memories and computing power in the same or smaller die area (following Moore's Law) also saw the first ARM Cortex M7 core product as well as the first low-power ARM Cortex M4 as a mix of low power consumption and power-processing performances. This suits perfectly the new wave of wearable products' requirements, according to Franck Martins, MCU marketing and application, senior manager, GC&SA region at ST Microelectronics.

However, Aravind Navada, senior engineering manager, Integrated Systems and Transceivers Group, Analog Devices, India, has a very different perspective. He says, "In my opinion, there is no such thing as an IoT-targeted MCU.

What's new

- New MCUs on 40nm process node
- Semiconductor firms moving up from silicon level to platform level
- New tools being delivered to lower software development and maintenance costs
- Tamper-protection and encryption features are coming with the chips. Some examples are BroadSAFE, SafeAssure and hardware safety modules
- Support for newer communication interfaces
- Better peripheral and sensor integration

Chip: PIC24FJ 1024GA610/GB610 family

Company: Microchip

Features:

- PIC24F 16-bit MCU featuring large dual-partition flash with live update capability
- Includes up to 1024kB flash, 32kB RAM, USB and advanced peripherals
- A combination of features makes it ideal for always-on applications and applications that require high memory

Architecture: 16-bit

